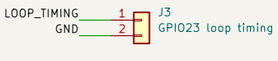
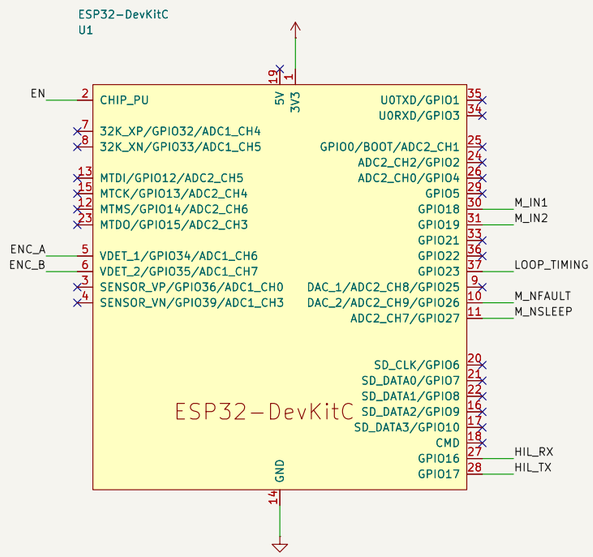
ESP32 DC motor speed controller — intended wiring (6 V N20 gearmotor)

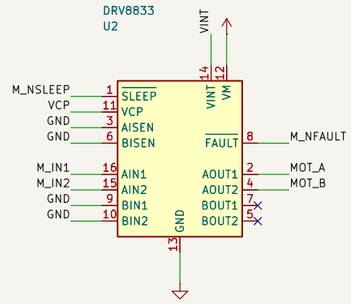
Pins are proposed. The firmware currently simulates the motor and uses only

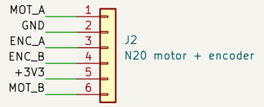
GPIO23 (loop timing), the UART0 console and the UART2 HIL link; the others are not driven.



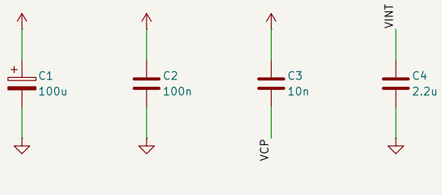














J2 pin order is typical for N20 hall-encoder motors: check yours before wiring.

J4 is the HIL link (UART2, GPIO17 TX / GPIO16 RX) to the Pi; the console stays on UART0 (USB).

ESP32 and Raspberry Pi are 3.3 V logic. The 6 V supply must share GND with both.

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